

Related Work / Survey of Previous Work [Date: 2026-05-11]

Goal

To evaluate selected open-source software projects that implement a "Modbus TCP gateway" or a "Modbus TCP-to-RTU bridge" on the ESP32 platform.

The study focuses on the software architecture, protocol support, maintainability, dependencies, framework compatibility, and development ecosystem of each project.

Selected Open-Source Projects

- <https://github.com/Rob2011/ESP32.Modbus-TCP-gateway>
 - <https://github.com/vwetter/esp32-modbus-gateway>
 - <https://github.com/zivillian/esp32-modbus-gateway>
 - <https://github.com/harihanv/esp32-modbus-gateway>
 - <https://github.com/tobiasfaust/SolaxModbusGateway>
 - https://github.com/rosenrot00/esphome_modbus_bridge
 - <https://github.com/eModbus/eModbus>
 - <https://github.com/NamNamIoT/ESP32>
 - <https://github.com/maxx-ukoo/esp32-modbus-tcp2rtu>
 - <https://github.com/espressif/esp-modbus>
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Guidelines for Comparison / Evaluation Criteria

Repository Maintenance and Activity

- Whether the project or GitHub repository is actively maintained and updated.
 - Last update or commit within 3 months
 - Last update or commit within 6 months
 - Last update or commit within 1 year
 - Last update or commit more than 1 year ago
- Development activity indicators:
 - Number of commits
 - Recent issue activity
 - Recent pull requests
 - Presence of active contributors

Project Popularity

- Popularity of the project based on GitHub statistics.
 - Number of stars
 - Number of forks

Software Framework and Platform Support

- Whether the project uses:
 - Arduino-ESP32 core
 - ESP-IDF framework
 - ESPHome
 - PlatformIO

- The version of the software framework (if used):
 - Arduino-ESP32 core
 - ESP-IDF
 - PlatformIO configuration
 - ESPHome
- Supported ESP32 platforms:
 - ESP32
 - ESP32-S2
 - ESP32-S3
 - ESP32-C3
 - ESP32-C6
 - ESP32-H2

Network Interface Support

- Whether the project supports:
 - Modbus TCP over WiFi only
 - Modbus TCP over Ethernet only
 - Both WiFi and Ethernet
- Whether the project supports Ethernet interfaces:
 - Internal Ethernet MAC + LAN8720
 - WS5100 / W5500
 - Other Ethernet modules

Modbus Protocol Features

- Whether the repository implements:
 - a Modbus TCP master/client library
 - a Modbus TCP slave/server library
 - a Modbus RTU master/client library
 - a Modbus RTU slave/server library
 - a Modbus ASCII implementation
 - a Modbus TCP-to-RTU gateway (bridge)
- Supported Modbus function codes and Modbus register types.
- Support for:
 - asynchronous communication
 - non-blocking operation
 - multi-client TCP connections
 - FreeRTOS task-based communication

Software Architecture and Implementation

- Software design characteristics:
 - synchronous or asynchronous architecture
 - use of FreeRTOS tasks
 - modular software structure
 - callback-based APIs
 - event-driven design
- Reliability-related features:
 - reconnect handling
 - watchdog handling
 - timeout handling
 - exception handling
 - CRC verification

Dependency Analysis

- Dependency on external libraries and their versions.
- Dependence on:
 - Arduino libraries
 - ESP-IDF components
 - AsyncTCP
 - WiFi libraries
 - Ethernet libraries
 - RS485 libraries
- Complexity of dependency management.

Build System and Development Tools

- Which software tools are used to build and test the firmware:
 - Arduino IDE
 - PlatformIO
 - ESP-IDF build system
 - CMake
 - ESP Component Manager
- Presence of:
 - GitHub Actions
 - CI/CD pipelines
 - automated testing
 - unit tests
 - example projects

Documentation Quality

- Availability and quality of:
 - README documentation
 - API documentation
 - example code
 - wiring diagrams
 - setup instructions
 - protocol descriptions

Practical Deployment Features

- Suitability for industrial or IoT deployment.
- Support for:
 - RS485 transceivers
 - UART direction control (DE/RE)
 - WiFi reconnection
 - Ethernet failover
 - OTA firmware update
 - static IP configuration
- Ease of deployment and configuration.
